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Verification Behaviour as a Link Between Trust in Augmented Analytics and Perceived Decision Quality

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Abstract: Augmented analytics shifts much of the analytical work in business intelligence (BI) to artificial intelligence (AI), leaving non-technical users with one remaining safeguard: verifying AI-generated insights before acting on them. This paper examines whether trust in augmented analytics is associated with less verification and how verification relates to perceived decision quality. Survey data from 350 non-technical BI users in Vietnam were analysed with partial least squares structural equation modelling (PLS-SEM), using a newly developed ten-item verification behaviour scale (VER). The results suggest that trust is negatively associated with verification, and verification is positively associated with perceived decision quality. The indirect association of trust through verification is negative, whereas its direct association remains positive, consistent with competitive mediation. Perceived explainability is associated with higher trust, but its direct association with verification is not significant, and the hypothesized moderating role of AI literacy receives no statistical support. Because the design is cross-sectional and self-reported, the paths represent associations rather than causal effects. Automation complacency and objective decision quality were not measured, and the verification scale was supported by initial validation evidence from one sample. The findings motivate linking insights to their source data, second checks for high-value decisions and direct monitoring of verification; the effectiveness of these proposals remains untested. The study develops a survey measure of verification behaviour with initial validation evidence and shows that the positive trust–decision quality association coexists with a negative indirect association through verification.

Keywords: Augmented analytics; Trust; Verification behaviour; Automation complacency; Perceived decision quality; PLS-SEM

1 Introduction

The increasing integration of artificial intelligence (AI) into business intelligence (BI) platforms has transformed how organizational users access and interpret analytical information. AI-generated insights, such as revenue trends, anomaly alerts, and forecasts, can provide decision-relevant information; however, users must still determine whether these outputs accurately reflect the underlying data and are appropriate for organizational use. This issue has become increasingly important as augmented analytics capabilities are embedded in BI platforms. These capabilities can respond to natural-language queries, detect anomalies, generate narrative summaries, and produce forecasts with limited support from specialist analysts [1]. Consequently, employees in sales, finance, operations, and human resources can perform analytical tasks that previously required substantial technical expertise or assistance from data analysts [2].

Recent research has increasingly positioned BI and AI as strategic organizational capabilities. BI capabilities have been linked to AI-enabled logistics innovation through improved decision support [3], and AI-enabled BI to digital marketing performance through data-driven decision-making [4]. Related studies have examined AI as a resource for improving managerial decision-making [5] and have considered its risks at institutional and governance

levels [6]. Less is known, however, about whether individual users verify AI-generated insights before acting on them, or how such verification behaviour relates to their perceived decision quality.

From a management perspective, the question concerns how organizations divide analytical work between people and AI. Raisch and Krakowski [7] distinguish automation, in which machines take over a task, from augmentation, in which humans and machines work on it together, and argue that the two are interdependent: tasks that begin as human–AI collaboration can shift towards automation over time, and an overemphasis on either approach can set off self-reinforcing cycles with harmful organizational consequences. Augmented analytics is typically introduced as augmentation, with the AI preparing insights and the employee remaining responsible for judging them. Verification is the point at which this division of labour is enacted in daily work. When users stop checking, an arrangement designed as augmentation may operate in practice as unacknowledged automation. Examining verification behaviour therefore links an individual-level mechanism to a managerial choice about how far AI-generated insights should be relied upon without review.

Existing research offers two contrasting perspectives on the role of trust in this verification process. Adoption-oriented research generally treats trust as a condition facilitating the acceptance and use of AI-generated recommendations, and the authors' earlier, non-peer-reviewed preprint applied this view to non-technical BI users [8].More broadly, trust is considered an important prerequisite for integrating AI into organizational routines and realizing its potential value [9]. In contrast, human-factors research suggests that high trust in automation can reduce users' motivation to monitor and verify system outputs, a pattern commonly described as automation complacency [10, 11]. Recent evidence has extended this concern to AI-based recommendations [12] and generative AI use at work, where confidence in AI tools has been associated with less critical evaluation of their outputs [13]. Accordingly, trust may relate to decision quality through two countervailing pathways: it may facilitate the adoption and use of AI-generated insights while simultaneously reducing verification and increasing the likelihood that inaccurate or contextually unsuitable outputs are accepted without scrutiny. This distinction is consistent with Schoeffer et al.'s [14] argument that reliance on AI should be conceptually separated from the quality of the decisions that result from that reliance.

Two research gaps limit current understanding of this relationship in organizational settings. First, existing evidence on AI overreliance has largely relied on experiments in which reliance is inferred from participants' agreement with intentionally correct or incorrect recommendations [15, 16]. Although such designs provide strong experimental control, they do not adequately capture how employees verify AI-generated outputs during routine BI use. Second, automation complacency has primarily been studied in controlled settings, including route-planning and medical-diagnosis tasks, rather than among non-technical employees using commercial BI platforms in their everyday work. It therefore remains unclear whether trust is associated with reduced verification in routine augmented-analytics use and whether verification behaviour helps explain the relationship between trust and perceived decision quality.

Accordingly, this study examines whether trust in augmented analytics is associated with less verification of AI-generated insights among non-technical users, whether verification behaviour mediates the relationship between trust and perceived decision quality, and how perceived explainability and AI literacy relate to verification behaviour, including whether AI literacy moderates the trust–verification relationship. These expectations are formalized in seven hypotheses (H1–H7), which are presented in Section 2.4.

To this end, this study analyses survey data from 350 non-technical BI users in Vietnam. Trust in the AI capabilities of BI software is negatively associated with self-reported verification behaviour ($\beta = -0.196$), whereas verification behaviour is positively associated with perceived decision quality ($\beta = 0.220$). Within the specified model, the resulting negative indirect association partially offsets the positive direct association between trust and perceived decision quality, so that the total association is approximately 14% smaller than the direct association. The direct association between perceived system explainability and verification is not statistically significant, and the hypothesized moderating role of AI literacy received no statistical support.

This study makes three contributions. First, it conceptualizes verification as a distinct form of user behaviour and develops a ten-item verification behaviour scale (VER) with initial validation evidence, extending experimental research on AI overreliance to organizational survey research. Second, it examines whether trust is associated with perceived decision quality through two pathways with opposing signs, thereby clarifying that the positive trust–outcome relationship assumed in BI-adoption research may represent a net association. Third, it examines the roles of perceived system explainability and AI literacy in verification behaviour, including whether AI literacy attenuates the negative association between trust and verification.

The present study differs from the authors' earlier non-peer-reviewed preprint [8] in its theoretical framing, analytical model and dataset. Whereas the preprint conceptualized trust as a uniformly beneficial component of an AI-adoption process grounded in cognitive delegation and analysed a sample of 250 respondents, the present study draws on automation-complacency theory to investigate whether trust may also be associated with reduced verification of AI-generated insights, and it analyses a separately collected sample of 350 respondents. The present study also introduces verification behaviour as a mediator between trust in augmented analytics and perceived

decision quality.

2 Literature Review and Hypotheses Development

2.1 Augmented Analytics and Non-Technical Users

Augmented analytics applies machine learning and natural language processing to automate data preparation, insight discovery and the explanation of results within BI platforms [1]. The organizational promise is self-service at scale: employees without analytical training can obtain answers without waiting for a specialist team. Research on self-service BI, however, indicates that such users frequently lack the expertise required to assess data scope, definitions and quality [2], and augmented features intensify this asymmetry. A narrative statement such as the one described in the introduction reaches the user already interpreted, so that the user's role shifts from producing an analysis to accepting or questioning one. Logg et al. [17] found that non-experts weighted algorithmic advice more heavily than equivalent human advice, a tendency the authors termed algorithm appreciation, whereas experts discounted it. In a BI context, this implies the possibility that users who rely heavily on augmented analytics may be less inclined to critically examine its outputs.

2.2 Trust Calibration and Automation Complacency

Lee and See [10] define trust in automation as the attitude that an agent will assist in achieving one's goals under conditions of uncertainty and vulnerability, and argue that favourable outcomes require calibrated trust, that is, trust commensurate with the actual capability of the system; overtrust can lead to misuse, whereas distrust can result in disuse. Parasuraman and Manzey [11] characterize the behavioural signature of overtrust as complacency, whereby operators reduce their monitoring and cross-checking of automated outputs, particularly when a system has demonstrated reliability and their attention is divided across competing demands. Two features of this account are relevant to the present study. Complacency is a matter of attention allocation rather than of ignorance, and has therefore been observed among experienced operators as well as novices. It may also remain undetected until the automated system fails, because a reliable system rarely penalizes omitted checks. Routine BI work satisfies both preconditions: most AI-generated summaries are correct on most occasions, and users consult them while engaged in other tasks. Evidence from generative AI points in the same direction. In a survey of knowledge workers, Lee et al. [13] found that higher confidence in AI tools was associated with lower self-reported critical thinking, whereas higher confidence in one's own abilities was associated with higher levels of critical thinking. Harbarth et al. [12] further demonstrated that attitudinal and behavioural dimensions of complacency may move in opposite directions: a transparent system reduced users' expectations of delegating tasks to it while increasing complacent behaviour. This finding cautions against treating trust and reduced checking as equivalent constructs.

2.3 Verification Behaviour

Verification behaviour is defined here as the frequency with which users check AI-generated insights against underlying data, the scope of the analysis, alternative sources, or input from other people before using them in reports, communications, or decisions. The construct is behavioural and is therefore distinct from trust, which is an attitude towards the system, and from perceived explainability, which is a perception of what the system discloses. It is also distinct from reliance, since a user may rely on an insight after verifying it, the pattern that Schoeffer et al. [14] identified as a condition for humans to improve upon AI recommendations. Following the scale-development procedure proposed by MacKenzie et al. [18], the construct domain was specified to cover four forms of checking: inspecting the data and scope behind an insight, recomputing or cross-checking its figures, consulting colleagues or data specialists, and assessing its plausibility against one's own knowledge.

2.4 Hypotheses

Figure 1 summarizes the research model. Self-confidence in one's own analytical skill, frequency of AI feature use, tenure with AI features and organizational function are included as controls on verification behaviour and perceived decision quality.

Explainability and trust. Lee and See [10] identify an understanding of how an automated system operates and the processes underlying its outputs as one of the three foundations of trust. A review of empirical work on trust in AI lists explainability among its most consistent antecedents [9], and a meta-analysis of 90 studies reports a significant but modest positive correlation between explainability and trust [19]. A competing argument holds that explanations may expose flaws in an AI's reasoning and thereby reduce trust. This argument is set aside in the present context because BI explanations largely describe the data scope and calculation behind routine summaries and rarely reveal reasoning errors to non-technical users.

H1: *Perceived explainability is positively associated with trust in augmented analytics.*

Explainability and verification. Explanations may also reduce the effort required to check an insight. Vasconcelos et al. [16] showed that explanations can reduce overreliance by lowering the cost of verifying AI-generated answers,

and Schemmer et al. [20], who defined appropriate reliance as the ability to distinguish correct from incorrect AI advice, demonstrated experimentally that explanations shape it. In a BI context, an explanation that names the period, region or data source behind a figure provides users with the material needed to check it. The competing mechanism is substitution, whereby users read the explanation in place of checking. Bansal et al. [15] found that explanations increased acceptance of AI recommendations irrespective of their correctness, and Harbarth et al. [12] observed more complacent behaviour with a transparent system. The positive prediction is retained because BI explanations direct attention to verifiable data rather than to model internals, although the substitution account is treated as the principal alternative.

H2: *Perceived explainability is positively associated with verification behaviour.*

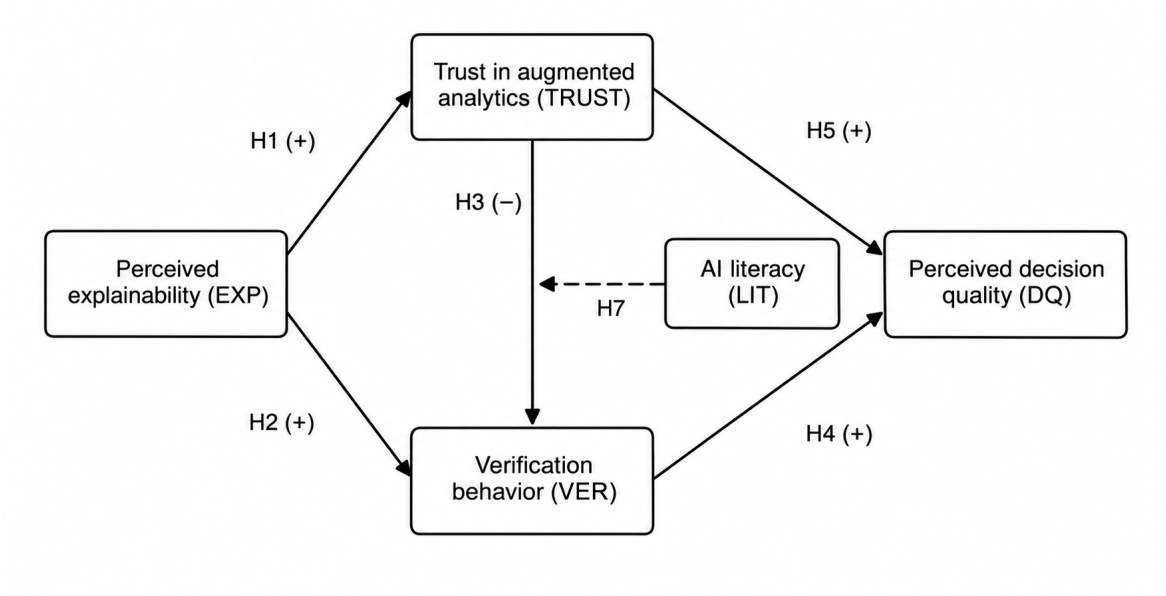


Figure 1. Research model

Note: Solid arrows denote hypothesized paths; the dashed arrow denotes moderation by AI literacy (H7). H6 is the indirect path TRUST → VER → DQ. AI = artificial intelligence.

Trust and verification. Complacency theory predicts that trust reduces the perceived need to monitor [11], and survey evidence links confidence in AI tools with less critical evaluation of their outputs [13]. A competing view holds that well-calibrated trust need not reduce checking, since users who trust a system based on its demonstrated reliability may still verify high-stakes outputs [10]. In routine BI use, however, trust is built largely from accumulated experience of a system that is usually correct, which is precisely the condition under which monitoring declines.

H3: *Trust in augmented analytics is negatively associated with verification behaviour.*

Verification and perceived decision quality. Verification can help identify errors in data scope, definitions or calculations before they enter a decision. Schoeffer et al. [14] show formally that human–AI complementarity depends on the ability to override incorrect advice, which requires users to evaluate the advice before relying on it. Visinescu et al. [21] also report that the quality of the information acted upon is associated with perceived decision quality in BI settings. Verification also carries costs, since it consumes time and may delay decisions. These costs are set aside here because the outcome of interest concerns how good and how successful decisions were judged to be rather than how quickly they were reached.

H4: *Verification behaviour is positively associated with perceived decision quality.*

Trust and perceived decision quality. This hypothesis tests an association that is widely assumed in adoption-oriented research and is included as scaffolding for the mediation test rather than as a novel claim. Users who trust a system act on its insights instead of disregarding them, thereby avoiding the losses associated with algorithm aversion [22], and the authors' earlier preprint proposed the same association for non-technical BI users [8], which the present study tests in an independent sample.

H5: *Trust in augmented analytics is positively associated with perceived decision quality.*

The mediating role of verification. If H3 and H4 hold, trust should also relate to perceived decision quality indirectly, through reduced verification, and that indirect association should be negative. Zhao et al. [23] classify a mediation pattern in which the direct and indirect effects have opposite signs as competitive mediation, whereby the indirect effect offsets part of the direct effect in the total association. This is the specific form that the complacency argument takes for decision outcomes [11].

H6: *Verification behaviour mediates the relationship between trust and perceived decision quality, such that the indirect association is negative.*

The moderating role of AI literacy. AI literacy refers to the competencies required to evaluate AI critically, to work with it and to use it as a tool [24]. The elaboration likelihood model holds that individuals scrutinize information when they possess both the motivation and the ability to do so [25]. Users with higher AI literacy may be better able to recognize where augmented analytics tends to fail and they may therefore be less likely to let trust reduce their verification efforts. The opposite expectation is also plausible, since users who feel competent with AI may become more comfortable delegating to it, and Parasuraman and Manzey [11] note that expertise does not necessarily protect against complacency. Because AI literacy is measured rather than manipulated, the hypothesis concerns a conditional association rather than a causal moderation.

H7: *AI literacy weakens the negative association between trust in augmented analytics and verification behaviour.*

3 Methodology

3.1 Sample and Procedure

The study employed a single cross-sectional survey of non-technical BI users working in organizations in Vietnam. Recruitment proceeded through four channels: occupational groups on Facebook (129 respondents, 36.9%), direct contact through a customer list (95, 27.1%), LinkedIn (67, 19.1%) and referral by earlier respondents (59, 16.9%). Responses were collected in a single survey round between 3 and 26 September 2026. The questionnaire defined BI software, AI features and "information produced by AI" using workplace examples, and four screening questions restricted participation to employed respondents working outside data, BI and information technology (IT) functions (A1–A2), who did not write code, build dashboards or data models, or operate statistical software themselves (A3), and who had used at least one built-in AI feature of their BI software in the preceding three months (A4). Respondents who had not made a decision or recommendation based on AI-generated information in that period (H0) were not required to complete the decision-quality section. Of the 428 questionnaires returned, 4 incomplete responses were removed, together with 8 completed in less than 120 seconds, 13 responses that failed an attention-check item embedded in the questionnaire (6 of which also contained identical answers to all ten verification items), 3 duplicate records and 50 respondents who answered "No" to H0. This left 350 valid responses, representing 81.8% of the returned questionnaires; the sample profile is summarized in Table 1.

Item-level missing values were rare (10 of the 11,900 responses across the 34 scale items, or 0.08%) and were replaced with the corresponding item means; nine were left blank and one was an out-of-range entry on EXP3.

The inverse square root method indicates that detecting a path coefficient of 0.15 with 80% power at a one-tailed 5% significance level requires 275 observations [26], so the sample of 350 exceeds the requirement for the main paths. It does not meet the requirement for interaction effects of the magnitude typically observed in survey research, a point to which Section 5.3 returns.

3.2 Measures

Unless stated otherwise, items used five-point agreement scales; full item wording and response anchors are provided in Table A1. Perceived explainability (EXP, five items) was adapted from the explanation satisfaction measures reviewed by Hoffman et al. [27] and reworded for AI-generated insights in BI software. Trust in augmented analytics (TRUST, seven items) was adapted from McKnight et al. [28], covering reliability (four items) and functionality (three items). AI literacy (LIT, three items) and perceived decision quality (DQ, four items) were adapted from Pinski and Benlian [29] and Visinescu et al. [21], respectively. DQ assessed respondents' evaluations of AI-informed decisions with known outcomes. VER was measured using a newly developed ten-item frequency scale.

The items were translated from English into Vietnamese and back-translated [30], then reviewed by two experts for clarity and relevance to BI work. Expert review prompted minor adjustments to BI terminology. The questionnaire was subsequently pre-tested with 30 users, following the sample-size recommendation of Perneger et al. [31]. All pre-test participants reported clear comprehension, and no further wording changes were made.

Control variables included self-confidence in analytical skill (single item), AI feature use frequency (five categories), AI feature tenure and organizational function. The latter was represented by four dummy variables, with sales and other functions as the reference group.

The VER scale covered the four checking domains defined in Section 2.3, with item assignments provided in Table A1. VER8, which concerns reading an insight's explanation, was assigned to data and scope checking because BI explanations describe the underlying data scope and calculations (Section 2.4).

VER was specified as reflective because its items were treated as interchangeable, covarying expressions of a general tendency to verify insights before use [18]. This specification is consistent with the complacency account underlying H3 [11]. A formative specification remains defensible because the items describe distinct checking acts (Section 5.3).

Table 1. Sample profile (n = 350)

Characteristic	Category	n (%)
Organizational function (A2)	Sales and other functions	100 (28.6)
	Operations, production, logistics	90 (25.7)
	Finance, accounting	59 (16.9)
	Human resources, administration	54 (15.4)
	Marketing	47 (13.4)
AI feature use, past 3 months (B1)	Daily	81 (23.1)
	2–4 times a week	110 (31.4)
	About once a week	64 (18.3)
	2–3 times a month	63 (18.0)
	Once a month or less	32 (9.1)
Tenure with AI features (B2)	Less than 1 month	0 (0.0)
	1–3 months	15 (4.3)
	3–6 months	72 (20.6)
	6–12 months	124 (35.4)
	More than 1 year	139 (39.7)
BI software used most (A5)	Power BI	210 (60.0)
	Tableau	63 (18.0)
	Looker	18 (5.1)
	Qlik	21 (6.0)
	Zoho Analytics	10 (2.9)
	Internal system	24 (6.9)
	Other	4 (1.1)
Position (I2)	Staff	221 (63.1)
	Team leader, supervisor	64 (18.3)
	Head or deputy head of department	55 (15.7)
	Director, executive board	10 (2.9)
Firm size, employees (I3)	Fewer than 50	75 (21.4)
	50 to 199	97 (27.7)
	200 to 999	119 (34.0)
	1,000 or more	59 (16.9)
Industry (I4)	Wholesale, retail, distribution	88 (25.1)
	Manufacturing	73 (20.9)
	Finance, banking, insurance	50 (14.3)
	Technology, telecommunications	42 (12.0)
	Energy	24 (6.9)
	Logistics, transport	34 (9.7)
	Other services	39 (11.1)
Age (I5)	Under 25	58 (16.6)
	25 to 34	121 (34.6)
	35 to 44	118 (33.7)
	45 and over	53 (15.1)
Gender (I6)	Male	181 (51.7)
	Female	165 (47.1)
	Other	4 (1.1)

Note: AI = artificial intelligence; BI = business intelligence.

3.3 Common Method Bias

To address common method bias from the single questionnaire [32], constructs appeared in separate blocks with individual instructions, verification used frequency rather than agreement anchors, decision quality was measured last, and responses were anonymous. Statistical checks included full collinearity variance inflation factors (VIFs) for the five focal constructs, with values above 3.3 indicating bias [33], and a marker variable model [34]. The four-item product retention tendency marker [35], a consumer trait theoretically unrelated to the study constructs, was placed between the trust and verification blocks.

3.4 Data Analysis

Partial least squares structural equation modelling (PLS-SEM) in SmartPLS 4 [36] was chosen for the prediction-oriented model, new scale, single-item controls and interaction term [37]. Estimation used path weighting, standardized data and mean replacement for missing values. Item skewness (-0.23 to 0.05) and excess kurtosis (-0.67 to -0.20) showed no marked non-normality; non-normal prediction errors motivated mean absolute error (MAE) comparisons with a linear model (LM).

Following confirmatory composite analysis [38], assessment covered loadings, internal consistency (Cronbach's α, ρ_A, ρ_C), convergent validity (average variance extracted, AVE) and discriminant validity (heterotrait–monotrait ratio, HTMT). Criteria were loadings ≥ 0.708, reliability 0.70–0.95, AVE ≥ 0.50, inner VIFs < 3 [37], and upper bounds of one-tailed HTMT bootstrap confidence intervals (CIs) below 0.85 [39].

Path tests used 10,000 bootstrap resamples, two-tailed percentile CIs and a fixed seed; mediation followed Zhao et al. [23]. H7 used the two-stage approach [40], with f^2 benchmarks of 0.005, 0.01 and 0.025 for small, medium and large effects [41]; conditional effects were estimated at AI literacy's mean $\pm$ 1 standard deviation.

Out-of-sample prediction was assessed with PLSpredict (10 folds, 10 repetitions, fixed seed, early-antecedent prediction) [42] and the cross-validated predictive ability test (CVPAT) [43]. Q^2_{predict} used the training-sample mean benchmark; item-level MAE comparisons used LM, and CVPAT used indicator-average (IA) and LM benchmarks. Two robustness checks re-estimated the model without controls or with the trust–verification path reversed; sensitivity analysis excluded VER8 and VER10.

4 Results

Table 2 summarizes the measurement model results.

Table 2. Measurement model assessment

Construct	Item	Loading	VIF	α	ρ_A	ρ_C	AVE
Perceived explainability (EXP)	EXP1	0.835	1.917	0.869	0.891	0.903	0.652
	EXP2	0.815	2.001	–	–	–	–
	EXP3	0.830	1.940	–	–	–	–
	EXP4	0.772	1.909	–	–	–	–
	EXP5	0.782	1.976	–	–	–	–
Trust in augmented analytics (TRUST)	TRE1	0.782	2.182	0.886	0.888	0.911	0.593
	TRE2	0.774	2.124	–	–	–	–
	TRE3	0.769	2.120	–	–	–	–
	TRE4	0.763	2.038	–	–	–	–
	TFU1	0.770	2.173	–	–	–	–
	TFU2	0.779	2.278	–	–	–	–
	TFU3	0.752	1.959	–	–	–	–
Verification behaviour (VER)	VER1	0.774	1.993	0.886	0.903	0.909	0.506
	VER2	0.813	2.364	–	–	–	–
	VER3	0.749	1.901	–	–	–	–
	VER4	0.756	1.893	–	–	–	–
	VER5	0.752	1.872	–	–	–	–
	VER6	0.737	1.793	–	–	–	–
	VER7	0.731	1.755	–	–	–	–
	VER8	0.499	1.261	–	–	–	–
	VER9	0.781	1.991	–	–	–	–
	VER10	0.410	1.160	–	–	–	–
AI literacy (LIT)	LIT1	0.856	1.866	0.815	0.821	0.890	0.729
	LIT2	0.865	1.751	–	–	–	–
	LIT3	0.841	1.792	–	–	–	–
Perceived decision quality (DQ)	DQ1	0.845	2.071	0.861	0.874	0.905	0.705
	DQ2	0.847	2.005	–	–	–	–
	DQ3	0.799	1.884	–	–	–	–
	DQ4	0.866	2.116	–	–	–	–

Note: All loadings were significant ($p < 0.001$; 10,000 bootstrap resamples). VIF = outer variance inflation factor; α = Cronbach's alpha; ρ_A and ρ_C = composite reliability; AVE = average variance extracted; AI = artificial intelligence. Reliability and AVE are reported once per construct; – = not applicable. Single-item controls are omitted.

The measurement model meets the established criteria, with one qualification concerning the new verification scale. Item loadings ranged from 0.731 to 0.866 for all items except two verification items, VER8 (0.499) and VER10 (0.410). Following Hair et al. [44], items loading between 0.40 and 0.70 are removed only where removal is required to bring reliability or AVE above threshold. Because VER already reached ρ_C = 0.909 and AVE = 0.506 with both items included, they were retained in order to preserve the content domain of the new scale, and they are examined further below and in Section 5.1. Cronbach's α ranged from 0.815 to 0.886, ρ_A from 0.821 to 0.903 and ρ_C from 0.890 to 0.911, and outer VIFs from 1.160 to 2.364.

Discriminant validity was established. All HTMT values among the focal constructs were at or below 0.328, and the upper bounds of the corresponding one-tailed 95% confidence intervals did not exceed 0.418 (Table 3). The highest HTMT value involving a control variable was 0.341, between VER and self-confidence. The Fornell–Larcker criterion was also satisfied (Table A4).

Table 3. Discriminant validity assessment

Construct	EXP	TRUST	VER	LIT	DQ
EXP	–				
TRUST	0.328 [0.418]	–			
VER	0.147 [0.229]	0.217 [0.314]	–		
LIT	0.122 [0.223]	0.054 [0.150]	0.284 [0.375]	–	
DQ	0.088 [0.193]	0.286 [0.380]	0.180 [0.274]	0.065 [0.169]	–

Note: Brackets show the upper bounds of one-tailed 95% bootstrap confidence intervals (10,000 resamples). HTMT = heterotrait–monotrait ratio; EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = artificial intelligence literacy; DQ = perceived decision quality. – = not applicable.

There is no evidence that common method variance poses a substantial threat to the results. Full collinearity VIFs for the five focal constructs, computed from the latent variable scores, ranged from 1.083 to 1.238. Correlations between the marker and the focal constructs ranged in absolute value from 0.008 to 0.092. In the marker model reported in Table A2, none of the marker's paths to TRUST, VER or DQ was significant (all $p \geq 0.186$), and the reported main-model path coefficients changed by at most 0.005 without altering any inference.

Turning to the structural model, inner VIFs ranged from 1.000 to 1.439 and the standardized root mean square residual (SRMR) was 0.059. The main-model R^2 values were 0.091 for TRUST, 0.200 for VER and 0.117 for DQ (Figure 2); the corresponding adjusted R^2 values were 0.089, 0.177 and 0.093, respectively.

Table 4 reports the hypothesis tests. Perceived explainability was positively associated with trust (β = 0.302, p < 0.001), supporting H1. Its direct association with verification was small, negative and not significant ($\beta = -0.071$, p = 0.191), so that H2 is not supported. Trust was negatively associated with verification ($\beta = -0.196$, p < 0.001) and verification was positively associated with perceived decision quality (β = 0.220, p < 0.001), supporting H3 and H4. Trust was also directly and positively associated with perceived decision quality (β = 0.304, p < 0.001), supporting H5.

Table 4. Hypothesis tests

Paths and Controls	Hypothesis	Coefficient (β)	t-Statistic	p-Value	95% CI	f^2	Decision
EXP → TRUST	H1	0.302	6.466	<0.001	[0.216, 0.400]	0.101	Supported
EXP → VER	H2	−0.071	1.307	0.191	[−0.175, 0.037]	0.006	Not supported
TRUST → VER	H3	−0.196	3.822	<0.001	[−0.297, −0.095]	0.043	Supported
VER → DQ	H4	0.220	3.787	<0.001	[0.107, 0.337]	0.046	Supported
TRUST → DQ	H5	0.304	6.189	<0.001	[0.211, 0.404]	0.098	Supported
TRUST → VER → DQ	H6	−0.043	2.464	0.014	[−0.083, −0.015]	–	Supported
LIT × TRUST → VER	H7	−0.076	1.467	0.142	[−0.173, 0.030]	0.007	Not supported

Note: H1–H6 are from the main model; H7 is from the moderation model. H6 is the specific indirect effect (competitive mediation). EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = artificial intelligence literacy; DQ = perceived decision quality; CI = confidence interval; f^2 = effect size; – = not applicable.

In the moderation model using the full ten-item VER scale, R^2 for VER was 0.205. The TRUST → VER coefficient in this model was −0.200. Inner VIFs ranged from 1.000 to 1.444, and the interaction term had a VIF of

1.031. The SRMR for the saturated model was 0.050.

In the main model, verification mediates a negative indirect association between trust and perceived decision quality (β = −0.043, 95% CI [−0.083, −0.015]). With the direct path positive and the indirect path negative, the pattern corresponds to competitive mediation [23], supporting H6.

Table 5 presents the additional paths, controls and effects. Among the controls in the main model, self-confidence in one's own analytical skill showed the strongest association with verification (β = 0.287, p < 0.001, f^2 = 0.094), whereas the associations of use frequency and AI feature tenure were not significant at the 5% level. Seven of the eight organizational-function coefficients had p-values above 0.05 and 95% confidence intervals that included zero (Table A6). The exception was marketing, which, relative to the reference group of sales and other functions, was associated with higher perceived decision quality (β = 0.285, p = 0.055); its percentile bootstrap confidence interval lay entirely above zero (95% CI [0.005, 0.581]). Because the p-value, which is based on the bootstrap standard error, and the percentile interval lead to different conclusions at the 5% level, this coefficient is reported as marginally significant.

Table 5. Additional structural model results

Paths and Controls	Coefficient (β)	t-Statistic	p-Value	95% CI	f^2
LIT → VER	0.174	3.499	<0.001	[0.081, 0.276]	0.035
Self-confidence → VER	0.287	5.921	<0.001	[0.190, 0.380]	0.094
Self-confidence → DQ	0.018	0.310	0.756	[−0.101, 0.132]	0.000
Use frequency → VER	0.072	1.424	0.154	[−0.025, 0.174]	0.006
Use frequency → DQ	−0.032	0.568	0.570	[−0.137, 0.080]	0.001
AI tenure → VER	0.059	1.161	0.246	[−0.039, 0.160]	0.004
AI tenure → DQ	−0.023	0.420	0.674	[−0.131, 0.088]	0.001
Function dummies → VER, DQ	−0.291 to 0.285	–	≥0.055	–	–
TRUST → DQ (total)	0.261	4.995	<0.001	[0.159, 0.363]	–
EXP → TRUST → VER	−0.059	3.132	0.002	[−0.102, −0.027]	–
EXP → VER (total)	−0.130	2.421	0.016	[−0.234, −0.025]	–
EXP → TRUST → VER → DQ	−0.013	2.180	0.029	[−0.027, −0.004]	–
TRUST → VER (LIT −1 SD)	−0.124	1.773	0.076	[−0.264, 0.008]	–
TRUST → VER (LIT +1 SD)	−0.276	3.631	<0.001	[−0.426, −0.124]	–
Index of moderated mediation	−0.017	1.272	0.203	[−0.046, 0.006]	–

Note: Conditional effects and the moderated-mediation index use the moderation model; other estimates use the main model. Self-confidence = confidence in analysing data without AI. Function-dummy coefficients: Table A6. AI = artificial intelligence; EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = AI literacy; DQ = perceived decision quality; CI = confidence interval; SD = standard deviation; f^2 = effect size; – = not applicable.

The total association of trust with perceived decision quality (β = 0.261, 95% CI [0.159, 0.363]) is therefore approximately 14% smaller than the direct association, so that, within the specified model, the negative indirect association through verification partially offsets the positive direct association. Explainability also related to verification through trust (β = −0.059, 95% CI [−0.102, −0.027]), and its total association with verification was negative and significant (β = −0.130, p = 0.016). The serial path from explainability through trust and verification to perceived decision quality was small but excluded zero (β = −0.013, 95% CI [−0.027, −0.004]).

Figure 2 summarizes the main-model estimates for H1–H6 and the moderation-model estimate for H7; controls are omitted for clarity. The interaction between AI literacy and trust on verification was not significant (β = −0.076, p = 0.142, 95% CI [−0.173, 0.030]) and its effect size was small (f^2 = 0.007), so that H7 is not supported. AI literacy itself was positively associated with verification (β = 0.165, p = 0.001 in the moderation model). The simple slopes ran counter to the hypothesized direction: the trust–verification association was −0.124 (p = 0.076) at one standard deviation below the mean of AI literacy and −0.276 (p < 0.001) at one standard deviation above. The difference between the slopes was not statistically reliable, however, and the index of moderated mediation also included zero (β = −0.017, 95% CI [−0.046, 0.006]).

Table 6 summarizes the predictive assessment, and Table A3 reports the item-level PLSpredict results. Q^2_{predict} was positive for TRUST (0.082) and VER (0.098) but negative for DQ (−0.031), so the model predicted DQ less accurately than the training-sample means. Because the prediction errors departed from normality, the LM comparison used MAE [42]: all seven trust items and seven of the ten verification items had lower prediction errors than the benchmark, indicating high and medium predictive power respectively. All four DQ items also had lower prediction errors than the LM benchmark, although their Q^2_{predict} values were negative. In CVPAT, the model outperformed the IA benchmark for TRUST and VER but not for DQ, and the LM benchmark for TRUST and DQ

but not for VER. The two benchmarks answer different questions: the training-sample mean shows whether the antecedents carry out-of-sample information about an outcome at all, whereas the LM shows whether the model's structure predicts as well as an unconstrained regression on the same items. Predictive support for DQ is therefore limited and depends on the benchmark: the model predicted DQ better than the LM benchmark but not better than the item means.

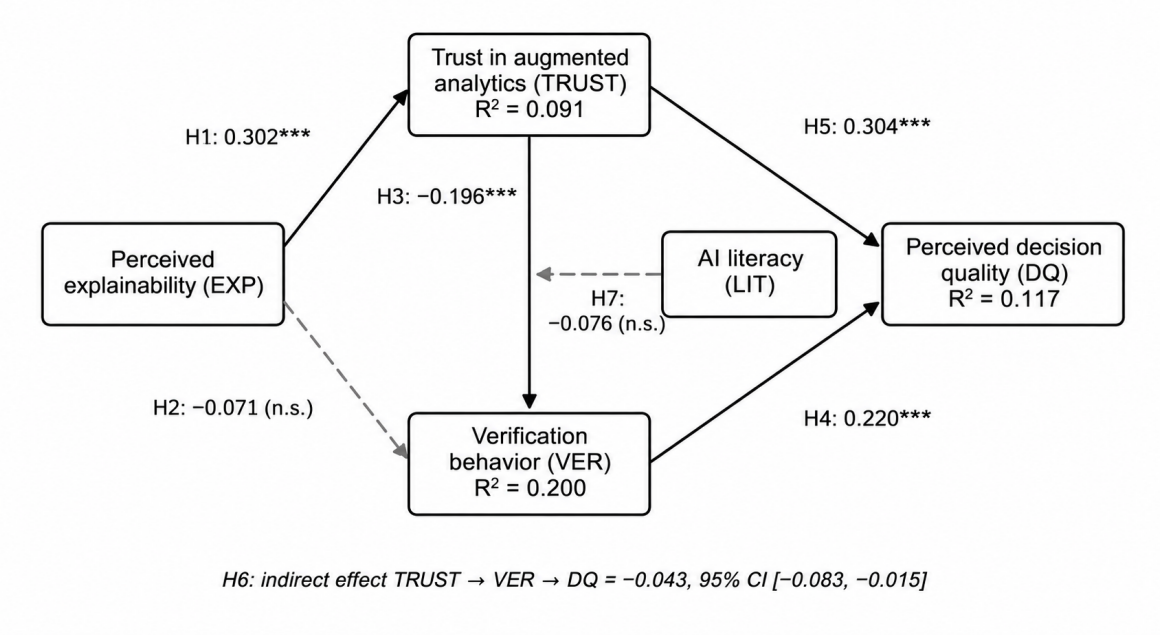


Figure 2. Estimated structural model

Note: *** denotes $p < 0.001$; n.s. = not significant; AI = artificial intelligence; CI = confidence interval.

Table 6. Predictive assessment by construct, benchmark and loss metric

Assessment	Benchmark	Loss Metric	TRUST	VER	DQ	Overall
PLSpredict: Q^2_{predict}	Training-sample mean of each item	Squared prediction error	0.082	0.098	−0.031	–
PLSpredict: item-level comparison	LM	MAE	7 of 7 items lower	7 of 10 items lower	4 of 4 items lower	–
CVPAT	IA	Average loss (squared prediction error)	Lower, $p = 0.011$	Lower, $p = 0.023$	Not lower, $p = 0.093$	Lower, $p = 0.006$
CVPAT	LM	Average loss (squared prediction error)	Lower, $p < 0.001$	Not lower, $p = 0.431$	Lower, $p < 0.001$	Lower, $p = 0.001$

Note: IA = indicator average; LM = linear model; MAE = mean absolute error; CVPAT = cross-validated predictive ability test; TRUST = trust in augmented analytics; VER = verification behaviour; DQ = perceived decision quality. CVPAT p-values test whether average loss is lower than the benchmark. Item-level PLSpredict results are in Table A3. – = not applicable.

The robustness checks answer different questions (Table A2). In the model without controls, the hypothesized paths retained their direction and statistical significance classification. The indirect association remained negative ($\beta = -0.042$, 95% CI [−0.080, −0.015]). The reversed model offers an alternative account of the same associations and does not confirm the direction of the proposed mediation: verification was negatively associated with trust ($\beta = -0.176$, $p = 0.001$). The direct explainability–verification path is sensitive to this choice. It was not significant in the main model ($\beta = -0.071$, $p = 0.191$) but was significant in the reversed model ($\beta = -0.136$, $p = 0.012$), close to its total association in the main model ($\beta = -0.130$). Among the other paths estimated in both the main and reversed models, none changed direction or statistical significance.

Table A7 compares the full ten-item scale with an eight-item scale excluding VER8 and VER10 in the main model with controls; H7 is estimated in the moderation model. Without the two items, the composite reliability of VER rose from 0.909 to 0.919 and its AVE from 0.506 to 0.587, while the trust–verification path, the verification–decision quality path and the specific indirect association kept their sign and significance and changed by at most 0.013. The direct explainability–verification path changed from -0.071 ($p = 0.191$) to -0.095 ($p = 0.050$).

Additional robustness checks with the eight-item scale (not tabulated) yielded a reversed VER $\rightarrow$ TRUST path of -0.175 ($p = 0.001$), an indirect TRUST $\rightarrow$ VER $\rightarrow$ DQ association of -0.039 ($p = 0.015$) without controls, and marker paths with $p \geq 0.200$. Both items are retained to preserve the content domain of the scale.

5 Discussion

Overall, trust in augmented analytics is associated with less verification of AI-generated insights (H3). Verification mediates a negative indirect association between trust and perceived decision quality that operates alongside a larger positive direct association (H4–H6). In the main model, perceived explainability had a negative indirect association with verification through trust and a negative total association, while the direct association was not statistically significant. AI literacy was positively associated with verification, but its hypothesized moderating role received no statistical support (H7).

5.1 Theoretical Implications

First, the negative trust–verification association is consistent with an automation-complacency explanation in knowledge work involving commercial BI software. Parasuraman and Manzey [11] associated complacency with reduced monitoring under conditions of high reliability and divided attention. The observed association is consistent with this explanation among users who were interpreting a narrative rather than monitoring an ongoing process. However, this association does not constitute direct evidence of complacency, as complacency was not measured. Moreover, because the reverse path was also statistically significant, the finding could equally reflect the possibility that users who engage in more verification tend to develop lower levels of trust. The cross-sectional design does not allow these alternative interpretations to be distinguished.

Second, the competitive mediation suggests that the positive trust–outcome association assumed in adoption-oriented research, including the authors' earlier preprint [8], may be interpreted as a net figure: within the specified model, the total association is approximately one seventh smaller than the direct association. Given the cross-sectional design and the use of perceived decision quality, these findings do not demonstrate that trust reduces decision quality by lowering verification. Nevertheless, this finding is consistent with the argument of Schoeffer et al. [14] that reliance and decision quality should be modelled separately, and with the distinction between calibrated trust and overtrust drawn by Lee and See [10]. From a management perspective, the pattern is consistent with the automation–augmentation paradox [7], in which tasks designed for human–AI collaboration can drift towards automation, but whether greater trust leads users to check less over time, and thereby weakens the human contribution to augmentation, cannot be determined from these data.

Third, in the main model, the direct association between perceived explainability and verification was not statistically significant, whereas both the indirect association through trust and the total association were negative. The negative indirect association is more consistent with the substitution account proposed by Bansal et al. [15] and Harbarth et al. [12] than with the cost-reduction account proposed by Vasconcelos et al. [16]. Because explainability was measured as a perceived construct rather than experimentally manipulated, and the direct path was sensitive to model specification (-0.071 to -0.136 across the models in Section 4), this does not show that explanations cannot encourage checking; whether the explanations in respondents' BI tools directed users to the data needed for checking was not examined here. AI literacy was associated with more verification, but its hypothesized moderation received no statistical support; the simple slopes pointed the other way, in line with the observation that expertise does not necessarily protect against complacency [11], but their difference was not statistically reliable. Self-confidence in one's own analytical skill was the strongest correlate of verification, echoing Lee et al. [13].

Fourth, the newly developed VER scale performed adequately but not uniformly, and its validation evidence comes from a single sample. VER8, which records reading the explanation displayed with an insight, loaded weakly (0.499), and its cross-loading on perceived explainability (0.273; Table A5) left the narrowest margin of any item, which suggests that, although the item was assigned to inspecting data and scope (Section 3.2), reading an explanation is closer to consuming the system's account than to checking it. VER10, which records assessing plausibility against one's own knowledge, describes a mental rather than an observable act. Excluding both items changed the H3, H4 and H6 estimates by at most 0.013 (Table A7). Both items are candidates for revision and validation in an independent sample.

5.2 Practical Implications

For managers, the results suggest that augmentation may need to be maintained by design rather than simply assumed [7]. The findings motivate two proposed actions and one test for organizations implementing augmented analytics. Their effectiveness in improving decision-making was not examined in this study and therefore requires evaluation in organizational practice. Figure 3 positions these steps along the pathway from an AI-generated insight to a decision and specifies, for each, the responsible actor, the trigger, the measure, and the supporting results.

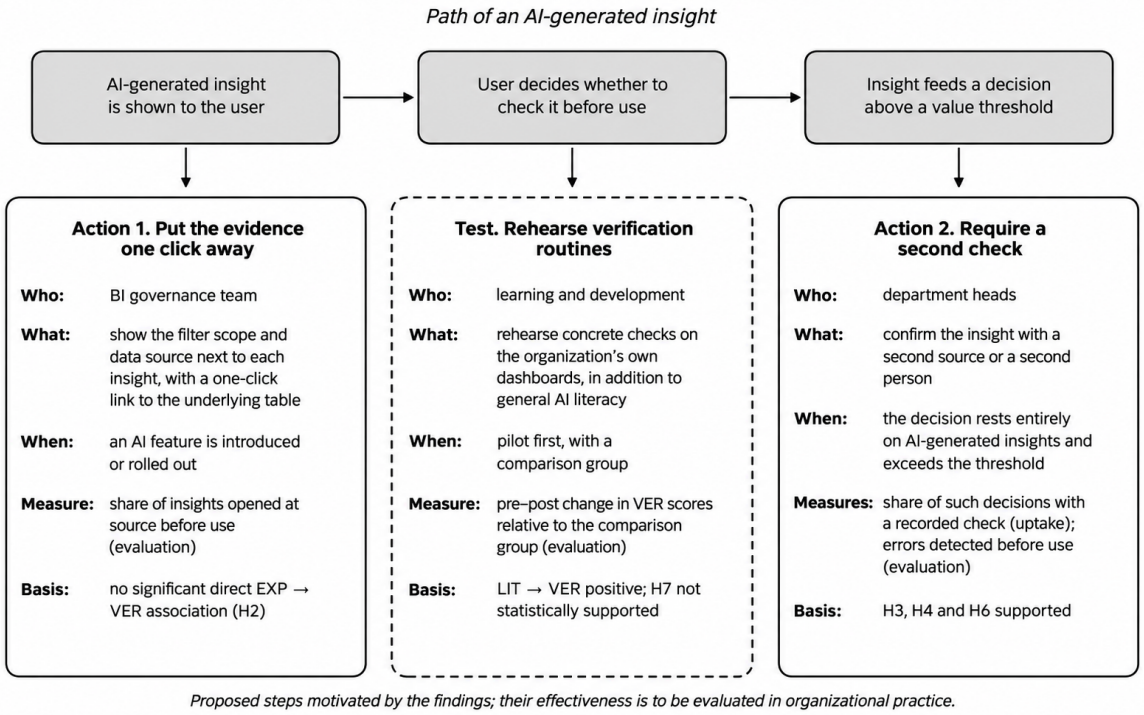


Figure 3. Where the three proposed steps sit along the path of an AI-generated insight

Note: Solid borders mark proposed actions; the dashed border marks a test to be piloted with a comparison group. Who acts, the trigger and the measures follow Section 5.2; Basis lists the results that motivate each step. AI = artificial intelligence; BI = business intelligence; EXP = perceived explainability; LIT = AI literacy; VER = verification behaviour.

Action 1: put the evidence one click away. Because perceived explainability showed no significant direct association with verification, the first proposal focuses on what explanations display rather than how many are provided. Whenever an AI feature is introduced or rolled out, BI governance teams can display the filter scope and data source alongside each generated insight, with a one-click route to the underlying table, and monitor the proportion of insights accessed at source before being used in a report or decision. Part of the time that AI saves in preparing analyses could be reallocated to this step.

Action 2: require a second check above a value threshold. Because trust was associated with less verification, while verification was associated with higher perceived decision quality, department heads can require confirmation from a second source or a second person for decisions based entirely on AI-generated insights that exceed a value threshold set by the organization. The proportion of such decisions with a recorded check shows whether the requirement is applied (uptake), and the number of errors detected before use shows whether the check identifies problems (evaluation). Such a requirement would act as a cognitive forcing function, which has been shown to reduce overreliance on AI in experimental settings [45].

Test: rehearse verification routines in addition to general AI literacy. AI literacy was positively associated with verification, but its hypothesized moderation (H7) received no statistical support, so the results neither support nor rule out general AI-literacy training. Learning and development teams could therefore pilot training that rehearses concrete verification routines using the organization's own dashboards with a comparison group, and assess its effectiveness through the pre–post change in VER scores relative to the comparison group rather than through test results.

5.3 Limitations and Future Research

The study adopts a cross-sectional design; therefore, the identified paths represent associations, and the reversed model indicates that the direction of the trust–verification relationship cannot be established from these data; automation complacency and objective decision outcomes were not measured. Decision quality is perceived and self-reported, and was collected in the same questionnaire as its predictors, so respondents with limited analytical expertise may misjudge their own decisions, while those who verify, or trust AI features, may rate their decisions more favourably. Predictive support for perceived decision quality is limited and depends on the benchmark used. A formative specification of VER is defensible because its items describe distinct acts, two items require revision, and the scale requires validation in an independent sample. The convenience sample of non-technical BI users in Vietnam, recruited through social-media groups, a customer list and referral, may differ from users in other settings in organizational culture and BI maturity, and excluding the 50 respondents who had not acted on AI-generated information may over-represent users who rely on AI output. Decision stakes, industry, system quality and organizational data culture were not measured and may partly account for the observed associations. AI literacy was measured rather than manipulated, and the study was not powered to detect a small interaction: applying the inverse square root method [26] to an interaction coefficient of the observed magnitude (0.076), with 80% power at a one-tailed 5% level, indicates that about 1,070 respondents would be required. Future research could use a two-wave design with later decision outcomes, scenario experiments with planted errors, system logs of verification and field evaluations of the proposed steps.

6 Conclusions

This study examined how trust in augmented analytics relates to non-technical BI users' verification of AI-generated insights and to their perceived decision quality. Trust was negatively associated with verification, while verification was positively associated with perceived decision quality, resulting in a negative indirect association alongside a larger positive direct association. In the main model, perceived explainability was positively associated with trust, whereas its direct association with verification was not statistically significant. AI literacy was associated with more frequent verification, but its hypothesized moderating role received no statistical support. These associations are consistent with automation complacency, but may also reflect users who develop lower trust as a result of more frequent verification. Predictive support for perceived decision quality depends on the benchmark used, and the verification scale has initial validation evidence only. The proposed steps remain to be evaluated in organizational practice, and the findings, based on non-technical BI users in Vietnam, may vary across organizational and technological contexts.

Author Contributions

Conceptualization and methodology, T.P.T.P. and H.N.M.; software, formal analysis and data curation, H.V.L.T.; investigation and validation, N.P.T.Y.; writing—original draft preparation, T.P.T.P.; writing—review and editing, N.P.T.Y. and H.N.M.; supervision and project administration, H.N.M. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent to participate was obtained from all participants on the questionnaire's first page. Participants were informed of the study's purpose, voluntary and anonymous participation, and their right to withdraw at any point.

Data Availability

The anonymised dataset and SmartPLS project file are available from the corresponding author upon request. The data contain no information identifying respondents or employers.

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This study builds on earlier work reported in a preprint [8], using a separately collected sample of 350 respondents and a different analytical model (see the Introduction and Section 5.1). Across the preprint's three versions, authorship and affiliations changed as the research scope widened, contributions changed and contributors moved institutions. The listed affiliations are those held during this study.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-Assisted Technologies

Claude (Anthropic) was used for language editing, reference formatting, and drafting selected passages and the graphical abstract. The authors reviewed and revised all content and take full responsibility for it.

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Appendix

A. Measurement Items and Supplementary Results

Table A1. Measurement items, descriptive statistics and loadings

Item	Item Wording (English Version)	Mean	SD	Loading
	Perceived explainability (EXP)			
EXP1 (C1)	I clearly know what the AI features of this software can (and cannot) help with when I need to decide something at work.	3.198	1.03	0.835
EXP2 (C2)	When the AI reaches a conclusion (e.g., "Revenue fell 12% in September"), I understand why it reached that conclusion.	3.206	0.99	0.815
EXP3 (C3)	I know which data the AI relies on to produce information.	3.169	1.01	0.830
EXP4 (C4)	This software tells me what the AI based the information on and how it was calculated.	3.114	1.02	0.772
EXP5 (C5)	The explanation this software provides is detailed enough for me to understand the information produced by the AI.	3.171	1.00	0.782
	Trust in augmented analytics (TRUST)			
TRE1 (D1)	The AI features of this software are reliable.	3.183	0.95	0.782
TRE2 (D2)	Whenever I need them, the AI features of this software help me.	3.169	1.07	0.774
TRE3 (D3)	I can safely rely on the AI features of this software.	3.221	1.05	0.769
TRE4 (D4)	The AI features of this software run stably when I use them (no freezing, errors or stopping midway).	3.109	1.06	0.763
TFU1 (D5)	This software has the AI features I need.	3.151	1.05	0.770
TFU2 (D6)	The AI features of this software meet the requirements of my work.	3.129	1.06	0.779
TFU3 (D7)	The AI features of this software do what I want them to do.	3.181	0.98	0.752
	Verification behaviour (VER)			
VER1 (F1)	Before using information produced by the AI, I open the detailed data table in this software to see the data the AI used.	2.805	1.17	0.774
VER2 (F2)	I re-check each figure stated in the AI's answer, summary or comment.	2.786	1.19	0.813
VER3 (F3)	Before using information produced by the AI, I check the data scope it was calculated on (period; region, branch, product group or cost item).	2.814	1.11	0.749
VER4 (F4)	I compare figures given by the AI with the same figures in sources outside this software (Excel files, accounting or sales software, earlier reports).	2.754	1.16	0.756

VER5 (F5)	When information produced by the AI differs from what I expected, I look for additional data to check it.	2.806	1.19	0.752
VER6 (F6)	When information produced by the AI seems odd, I ask colleagues, superiors or my staff before using it.	2.766	1.22	0.737
VER7 (F7)	When using AI-produced information for important matters (reports to superiors, major decisions), I ask someone knowledgeable about data (data or IT unit, analysts) to check it first.	2.789	1.18	0.731
VER8 (F8)	Before using information produced by the AI, I read the explanation the AI displays with it.	2.851	1.16	0.499
VER9 (F9)	Before using information produced by the AI, I check which source the data came from (e.g., which sales or accounting software or Excel file).	2.871	1.16	0.781
VER10 (F10)	Before using information produced by the AI, I consider whether it is plausible given what I know about the business situation.	2.885	1.22	0.410
	AI literacy (LIT)			
LIT1 (G1)	I know which tasks should be given to AI and which tasks people should do themselves.	3.234	1.04	0.856
LIT2 (G2)	In general, I understand how AI tools (such as ChatGPT, Copilot or AI in reporting software) work to produce results.	3.137	0.99	0.865
LIT3 (G3)	Considering all my experience, I am proficient in using AI.	3.186	1.01	0.841
	Self-confidence in own analytical skill (SC)			
SC1 (G4)	Without AI features, how confident are you that you could analyze data for your work yourself (e.g., read reports, compare periods, find out why figures rise or fall)?	3.175	1.06	–
	Perceived decision quality (DQ)			
DQ1 (H1)	Looking back, I think those were good decisions.	3.171	1.03	0.845
DQ2 (H2)	Overall, I am satisfied with the results those decisions brought.	3.194	1.05	0.847
DQ3 (H3)	I believe I chose the right option in those decisions.	3.206	1.02	0.799
DQ4 (H4)	Overall, those decisions achieved the results I wanted.	3.174	1.01	0.866
	Product retention tendency (MARKER)			
MK1 (E1)	I find it hard to get rid of my belongings.	3.140	1.03	0.726
MK2 (E2)	I tend to keep old things even if I rarely use them.	3.234	1.03	0.899
MK3 (E3)	I do not like throwing my belongings away.	3.131	1.08	0.825
MK4 (E4)	Unless I really have to get rid of something, I usually keep it rather than throw it away.	3.223	1.03	0.828

Note: Unless stated otherwise, items use five-point agreement scales (1 = strongly disagree, 5 = strongly agree). VER uses frequency anchors (1 = never, 5 = always); SC is a single-item control (1 = not at all confident, 5 = extremely confident). Adapted measures: EXP [27], TRUST [28], LIT [29] and DQ [21]. MARKER is from Haws et al. [35]; VER was developed for this study. VER domains: data and scope (items VER1, VER3, VER8, VER9), recomputing or cross-referencing (items VER2, VER4, VER5), consulting others (items VER6, VER7), plausibility (item VER10). DQ concerns decisions with known outcomes. AI = artificial intelligence; IT = information technology; SD = standard deviation; – = not applicable.

Table A2. Robustness checks

Path	Main Model	Without Controls	With Marker	Reversed Path
EXP → TRUST	0.302 (<0.001)	0.302 (<0.001)	0.300 (<0.001)	0.289 (<0.001)
EXP → VER	−0.071 (0.191)	−0.031 (0.638)	−0.069 (0.202)	−0.136 (0.012)
TRUST → VER	−0.196 (<0.001)	−0.196 (<0.001)	−0.191 (<0.001)	–
VER → TRUST	–	–	–	−0.176 (0.001)
VER → DQ	0.220 (<0.001)	0.212 (< 0.001)	0.225 (< 0.001)	0.219 (<0.001)
TRUST → DQ	0.304 (<0.001)	0.292 (<0.001)	0.300 (< 0.001)	0.305 (<0.001)
LIT → VER	0.174 (<0.001)	0.259 (<0.001)	0.174 (< 0.001)	0.172 (0.001)
LIT × TRUST → VER	−0.076 (0.142)	−0.087 (0.103)	–	–
TRUST → VER → DQ (indirect)	−0.043 (0.014)	−0.042 (0.013)	−0.043 (0.014)	–
MARKER → TRUST	–	–	0.080 (0.186)	–
MARKER → VER	–	–	−0.051 (0.342)	–
MARKER → DQ	–	–	0.054 (0.382)	–

Note: All models use the full ten-item VER scale. Coefficients (p-values in parentheses); interaction estimates are from the moderation models. EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = artificial intelligence literacy; DQ = perceived decision quality; MARKER = product retention tendency; – = not estimated or not applicable.

Table A3. Item-level PLSpredict results

Item	Q^2_{predict}	MAE (PLS-SEM)	MAE (LM)	MAE (PLS-SEM−LM)
TRE1	0.037	0.758	0.775	−0.017
TRE2	0.048	0.830	0.848	−0.018
TRE3	0.014	0.854	0.882	−0.027
TRE4	0.036	0.851	0.882	−0.030
TFU1	0.076	0.801	0.836	−0.035
TFU2	0.063	0.836	0.863	−0.027
TFU3	0.057	0.760	0.775	−0.015
VER1	0.083	0.926	0.931	−0.005
VER2	0.044	0.954	0.951	0.002
VER3	0.029	0.876	0.887	−0.011
VER4	0.077	0.911	0.919	−0.008
VER5	0.059	0.967	0.965	0.001
VER6	0.059	0.978	0.983	−0.005
VER7	0.056	0.951	0.970	−0.020
VER8	−0.004	0.982	0.955	0.027
VER9	0.075	0.931	0.936	−0.005
VER10	0.012	0.988	1.005	−0.017
DQ1	−0.026	0.829	0.838	−0.009
DQ2	−0.021	0.859	0.874	−0.014
DQ3	−0.026	0.828	0.837	−0.009
DQ4	−0.015	0.805	0.810	−0.005

Note: PLS-SEM = partial least squares structural equation modelling; MAE = mean absolute error; LM = linear model; VER = verification behaviour; DQ = perceived decision quality. Rounding may prevent displayed MAEs and their differences from reconciling exactly.

Table A4. Descriptive statistics, construct correlations and the Fornell–Larcker criterion

Construct	Mean	SD	EXP	TRUST	VER	LIT	DQ	SC
Perceived explainability (EXP)	3.172	0.82	0.807					
Trust (TRUST)	3.163	0.79	0.302	0.770				
Verification behaviour (VER)	2.813	0.82	−0.070	−0.197	0.711			
AI literacy (LIT)	3.186	0.87	0.105	0.039	0.243	0.854		
Perceived decision quality (DQ)	3.186	0.86	0.079	0.251	0.154	0.052	0.839	
Self-confidence (SC, single item)	3.175	1.06	0.103	−0.002	0.327	0.244	0.087	1.000

Note: SD = standard deviation; AI = artificial intelligence; TRUST = trust in augmented analytics; SC = self-confidence in own analytical skill. Diagonal entries are square roots of average variance extracted; off-diagonal entries are construct correlations.

Table A5. Cross loadings

Item	EXP	TRUST	VER	LIT	DQ	SC
EXP1	0.835	0.296	−0.095	0.104	0.077	0.112
EXP2	0.815	0.244	−0.018	0.069	0.076	0.043
EXP3	0.830	0.269	−0.105	0.082	0.075	0.078
EXP4	0.772	0.179	0.001	0.067	0.049	0.085
EXP5	0.782	0.193	−0.025	0.097	0.026	0.096
TRE1	0.209	0.782	−0.107	0.029	0.226	−0.049
TRE2	0.231	0.774	−0.162	0.024	0.194	−0.029
TRE3	0.157	0.769	−0.168	0.063	0.199	−0.004
TRE4	0.205	0.763	−0.114	−0.009	0.250	0.016
TFU1	0.293	0.770	−0.174	0.013	0.197	0.039
TFU2	0.265	0.779	−0.161	0.070	0.171	0.029
TFU3	0.251	0.752	−0.172	0.028	0.113	−0.022
VER1	−0.061	−0.173	0.774	0.194	0.116	0.293
VER2	−0.050	−0.175	0.813	0.171	0.056	0.228
VER3	−0.102	−0.131	0.749	0.214	0.132	0.180
VER4	−0.072	−0.165	0.756	0.226	0.099	0.251
VER5	−0.084	−0.162	0.752	0.159	0.086	0.266
VER6	−0.069	−0.117	0.737	0.125	0.124	0.272
VER7	−0.063	−0.105	0.731	0.149	0.160	0.245
VER8	0.273	−0.011	0.499	0.201	0.180	0.148
VER9	−0.077	−0.201	0.781	0.211	0.096	0.264
VER10	−0.091	−0.098	0.410	0.054	0.081	0.107
LIT1	0.056	0.050	0.199	0.856	0.035	0.174
LIT2	0.110	0.022	0.228	0.865	0.062	0.250
LIT3	0.100	0.031	0.192	0.841	0.033	0.195
DQ1	0.042	0.199	0.142	0.023	0.845	0.094
DQ2	0.053	0.183	0.185	0.050	0.847	0.073
DQ3	0.076	0.209	0.066	0.073	0.799	0.044
DQ4	0.093	0.249	0.111	0.035	0.866	0.077
SC1	0.103	−0.002	0.327	0.244	0.087	1.000

Note: EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = artificial intelligence literacy; DQ = perceived decision quality; SC = self-confidence in own analytical skill.

Table A6. Coefficients of the organizational-function dummies (reference group: sales and other functions)

Path	Coefficient (β)	SE	t-Statistic	p-Value	95% CI
Marketing → VER	−0.276	0.156	1.776	0.076	[−0.573, 0.034]
Human resources, administration → VER	−0.152	0.151	1.008	0.313	[−0.446, 0.143]
Finance, accounting → VER	−0.291	0.158	1.839	0.066	[−0.601, 0.018]
Operations, production, logistics → VER	−0.117	0.139	0.842	0.400	[−0.381, 0.162]
Marketing → DQ	0.285	0.149	1.922	0.055	[0.005, 0.581]
Human resources, administration → DQ	0.039	0.171	0.227	0.821	[−0.292, 0.378]
Finance, accounting → DQ	0.170	0.156	1.089	0.276	[−0.140, 0.478]
Operations, production, logistics → DQ	0.110	0.148	0.741	0.459	[−0.177, 0.406]

Note: VER = verification behaviour; DQ = perceived decision quality; SE = standard error; CI = confidence interval.

Table A7. Sensitivity of the results to the exclusion of VER8 and VER10

Measure	Full Scale (10 Items)	Excluding VER8 and VER10 (8 Items)
VER composite reliability (ρ_C)	0.909	0.919
VER average variance extracted	0.506	0.587
R^2 (VER)	0.200	0.205
R^2 (DQ)	0.117	0.112
EXP → VER (H2)	−0.071 (p = 0.191)	−0.095 (p = 0.050)
TRUST → VER (H3)	−0.196 (p < 0.001)	−0.194 (p < 0.001)
VER → DQ (H4)	0.220 (p < 0.001)	0.207 (p < 0.001)
TRUST → DQ (H5)	0.304 (p < 0.001)	0.302 (p < 0.001)
TRUST → VER → DQ (H6), 95% CI	−0.043 [−0.083, −0.015]	−0.040 [−0.078, −0.014]
LIT × TRUST → VER (H7), f^2	−0.076 (0.007)	−0.070 (0.006)
Q^2_{predict} (DQ)	−0.031	−0.031

Note: H7 coefficients (f^2 in parentheses) are from the moderation model; other estimates use the main model with controls. Bootstrap: 10,000 resamples, two-tailed percentile CIs, same fixed seed. EXP = perceived explainability; TRUST = trust in augmented analytics; VER = verification behaviour; LIT = artificial intelligence literacy; DQ = perceived decision quality; CI = confidence interval; f^2 = effect size.